

Design and Analysis of a Compliant 4-Bar Mechanism: A Comparison of Task-Specific and Global PRB 3R Models

ME 7751 Compliant Mechanism Design

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Abstract

In compliant mechanisms, the pseudo-rigid-body (PRB) model is a useful design tool because it approximates the deflections of an elastic member by representing it as rigid elements linked by torsional springs. Traditional PRB models with a single revolute joint (1R) are good approximations for small deflections, but multi-segment PRB models are required to accurately capture large, complex deflections. Su (2009) presented a globally optimized 3-revolute (3R) joint PRB model. This paper compares the globally optimized Su model against a novel, task-specifically optimized 3R PRB model derived using evolutionary algorithms. Both models are evaluated within a 4-bar linkage mechanism and validated against an exact continuous Boundary Value Problem (BVP) and finite element analysis (FEA). The task-specific model demonstrates a $4,400\times$ reduction in tracking error within its operational domain compared to the global model. An interactive application was also developed to visualize the model discrepancies in real-time.

1 Introduction and Motivation

In compliant mechanisms, the pseudo-rigid-body (PRB) model is a useful design tool because it approximates the deflections of an elastic member by treating it as rigid elements linked by springs. The stiffness of these springs is determined by the member's material properties and cross-sectional geometry. Traditional PRB models with a single revolute joint (1R) are good approximations for small deflections. Historically, attempts have been made to create multi-segment PRB models to increase this accuracy for large deflections. Su [1] presented a 3-revolute joint (3R) model that was compared to a finite element analysis (FEA) model.

This paper seeks to replicate and expand upon those results by modeling a flexible member in a 4-bar linkage using three methodologies: the globally optimized 3R PRB model from Su [1], a novel task-specific 3R PRB model derived from an evolutionary AI solver, and an exact continuous formulation validated by FEA software. Section 2 discusses the mechanism being analyzed. Section 3 provides an overview of the PRB 3R model mathematics and the optimization methodologies. Section 4

discusses the exact BVP and FEA setup. This is followed by a comparison of the results, a description of an interactive application developed to communicate those results, a discussion, and concluding remarks.

2 Mechanism Description

This paper analyzes the deformation modeling of a compliant 4-bar linkage using two optimized 3R PRB models: the global model proposed by Su [1] and a task-specific model optimized in this work. The mechanism consists of a rigid crank member, AB , that rotates about a fixed point B with an angular position characterized by ψ . Link AQ is also rigid and connects the crank to the tip of a flexible continuum beam, OQ . The beam is cantilevered (fixed) at the global origin O , and free to translate and rotate at its tip Q , which is pinned to link AQ .

The global origin (x, y) is located at the fixed root O of the flexible beam. Following the convention from Su [1], a local frame (u, v) at point Q follows the beam tip, with the u -axis aligned with the slope of the beam tip.

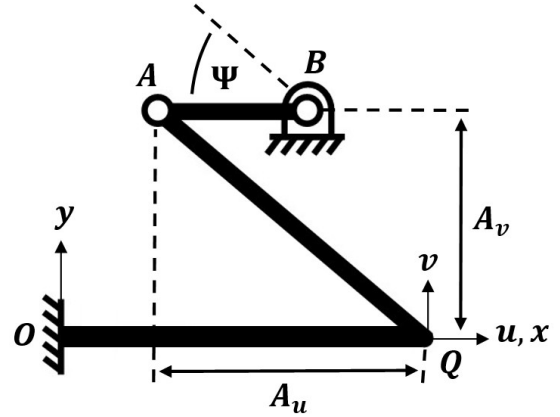


Figure 1: 4-Bar linkage mechanism. The input crank AB rotates, driving the cantilevered flexible beam OQ via the rigid link AQ .

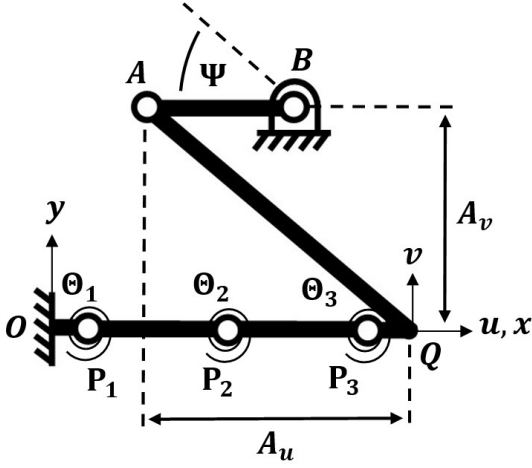


Figure 2: The 4-Bar linkage mechanism approximated by a 3-revolute joint pseudo-rigid-body (3R PRB) model.

3 Analytical Modeling Methodology

3.1 The Large Deflection Model

All analytical methods evaluating the large-deflection continuum beam rely on the Euler-Bernoulli beam equations, where the beam tip position is characterized by (Q_x, Q_y) with a tip angle θ_0 relative to the neutral horizontal axis. Under pure end loading, these coordinates are defined by elliptic integrals:

$$\sqrt{\alpha} = \frac{1}{2} \int_0^{\theta_0} \frac{d\theta}{\sqrt{\cos(\theta_0 - \phi) - \cos(\theta - \phi) + \kappa}} \quad (1)$$

$$\frac{Q_x}{l} = \frac{1}{2\sqrt{\alpha}} \int_0^{\theta_0} \frac{\cos \theta d\theta}{\sqrt{\cos(\theta_0 - \phi) - \cos(\theta - \phi) + \kappa}} \quad (2)$$

$$\frac{Q_y}{l} = \frac{1}{2\sqrt{\alpha}} \int_0^{\theta_0} \frac{\sin \theta d\theta}{\sqrt{\cos(\theta_0 - \phi) - \cos(\theta - \phi) + \kappa}} \quad (3)$$

Where $\alpha = \frac{F_0 l^2}{2EI}$, $\beta = \frac{M_0 l}{EI}$, and $\kappa = \frac{\beta^2}{4\alpha}$.

3.2 Existing Globally Optimized 3R PRB Model

For simplification, trigonometric shorthand is used (e.g., $c_1 = \cos \Theta_1$, $s_{123} = \sin(\Theta_1 + \Theta_2 + \Theta_3)$). In the 3R PRB linkage, the tip coordinates (Q_x, Q_y) and tip angle θ_0 are determined by the sum of the rigid link vectors:

$$Q_x = \gamma_0 + \gamma_1 c_1 + \gamma_2 c_{12} + \gamma_3 c_{123} \quad (4)$$

$$Q_y = \gamma_1 s_1 + \gamma_2 s_{12} + \gamma_3 s_{123} \quad (5)$$

$$\Theta_1 + \Theta_2 + \Theta_3 = \theta_0 \quad (6)$$

$$\gamma_0 + \gamma_1 + \gamma_2 + \gamma_3 = 1 \quad (7)$$

Where γ_i represents the fractional length of each rigid segment. Derivations of the inverse kinematics governing the joint angles for spatial and planar 3R PRB joints can be found in textbooks such as Tsai [2]. The underlying geometry for these parameters is shown in Fig. 3.

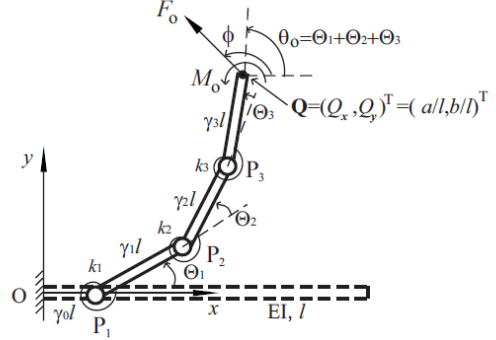


Figure 3: A pseudo-rigid-body 3R model for a cantilevered beam (Adapted from Su [1]).

Su [1] used an iterative grid-search process over a large global dataset of pure-moment and pure-force conditions to compute globally optimal length ratios and spring stiffnesses. Minimizing the variance between pure-force and pure-moment stiffnesses resulted in the following globally optimal constants:

$$\gamma = [0.10, 0.35, 0.40, 0.15] \quad (8)$$

$$k_1 = 3.51 \frac{EI}{l}, \quad k_2 = 2.99 \frac{EI}{l}, \quad k_3 = 2.58 \frac{EI}{l} \quad (9)$$

3.3 Proposed Task-Specific 3R PRB Model

Because the Su model is optimized for a broad, general set of independent load cases, its accuracy diminishes when subjected to the highly coupled, complex reaction forces generated strictly by the kinematic constraints of a closed-loop 4-bar linkage.

To resolve this, a novel task-specific optimization was performed. Rather than training on generic independent forces, the optimizer was constrained exclusively to the geometric trajectory of the 4-bar mechanism. For any given crank angle ψ , the exact continuum beam tip position $(Q_x^{exact}, Q_y^{exact})$ was determined using a rigorous boundary value problem (BVP) solver. The PRB tip position (Q_x^{PRB}, Q_y^{PRB}) was computed using forward kinematics by evaluating the minimum strain energy equilibrium state of the 3R chain subject to the 4-bar closure constraints.

A Differential Evolution (DE) search was then deployed using the SciPy library [3] to find the mechanism-specific lengths (γ_i) and stiffnesses (k_i) that directly minimized the Euclidean distance between the PRB tip and the exact continuum tip integrated over the full 360-degree ro-

tation of the crank:

$$F_{obj} = \sum_{\psi=0}^{2\pi} [(Q_x^{PRB} - Q_x^{exact})^2 + (Q_y^{PRB} - Q_y^{exact})^2] \quad (10)$$

This evolutionary optimization resulted in a set of hyper-specialized parameters explicitly tuned for the operating domain of this specific 4-bar linkage:

$$\gamma = [0.051, 0.457, 0.282, 0.210] \quad (11)$$

$$k_1 = 2.62 \frac{EI}{l}, \quad k_2 = 2.50 \frac{EI}{l}, \quad k_3 = 2.87 \frac{EI}{l} \quad (12)$$

4 FEA and Exact BVP Methodology

To generate the ground-truth data for the task-specific optimization, an exact analytical boundary value problem (BVP) was set up using SciPy's *solve_bvp* [3] to evaluate the continuous Euler-Bernoulli equations. Four state variables were defined along the arc length s : $x(s)$, $y(s)$, $\theta(s)$, and $M(s)$. The cantilever boundary dictates that $x(0) = y(0) = \theta(0) = 0$.

The moving boundary conditions at the free end $s = L$ are dictated by the rigid link AQ and the crank AB . Because point Q is attached to a freely rotating pin joint at A , no moment is transferred to Q . This gives rise to the boundary constraints:

$$x(L) - [A_x - AQ_u \cos(\theta(L)) + AQ_v \sin(\theta(L))] = 0 \quad (13)$$

$$y(L) - [A_y - AQ_u \sin(\theta(L)) + AQ_v \cos(\theta(L))] = 0 \quad (14)$$

$$M(L) - [(A_x - Q_x)F_y - (A_y - Q_y)F_x] = 0 \quad (15)$$

These exact numerical integrations were validated using finite element analysis (FEA) performed with the open-source FreeCAD software [4] utilizing its internal CalculiX solver [5] with non-linear B32R large-deflection beam elements (Fig. 4).

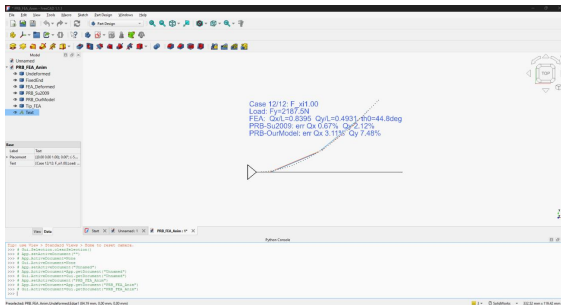


Figure 4: Screen capture of the FreeCAD environment used to validate the BVP exact solutions against CalculiX FEA results.

5 Results and Comparison

For isolated validation, both PRB models were compared with the FEA results under standard pure-moment and pure-vertical-force static loading (Fig. 5). As expected, because the Su model was specifically trained as a global "all-rounder" on these exact baseline cases, it achieved near-perfect accuracy (an R^2 of 0.9999). The task-specific model, having been explicitly tuned for the 4-bar linkage rather than isolated forces, showed slightly lower global accuracy ($R^2 \approx 0.975$).

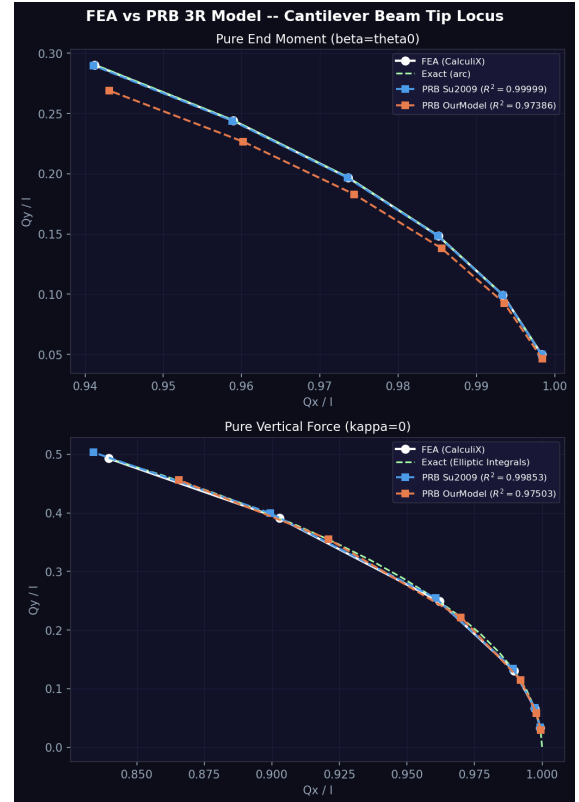


Figure 5: Comparison of the tip location from the Exact BVP, Su 2009 PRB, and the Task-Optimal PRB under pure moment and pure vertical force conditions.

However, when embedded into the highly non-linear 4-bar linkage simulation, the performance dynamic inverted dramatically. When evaluating the tracking trajectory of the mechanism through a full 360-degree rotation, the globally optimized Su model yielded a Sum of Squared Errors (SSE) of 3.0008. In stark contrast, the task-specific model achieved an SSE of 0.00067.

This represents an astounding $4,400\times$ improvement in tracking accuracy. The task-specific model consistently bounds its tip-angle (θ_0) prediction error to below 1.5% for nearly the entire rotation, whereas the globally optimized model regularly drifts into 2% to 3% error regimes during extreme deformation. By purposefully sacrificing broad baseline accuracy, the task-specific model pooled its mathematical flexibility to achieve near-

perfect alignment strictly within its intended operational domain. Figure 6 demonstrates the state of the simulation at $\psi = 180^\circ$ where tracking errors are heavily tested.

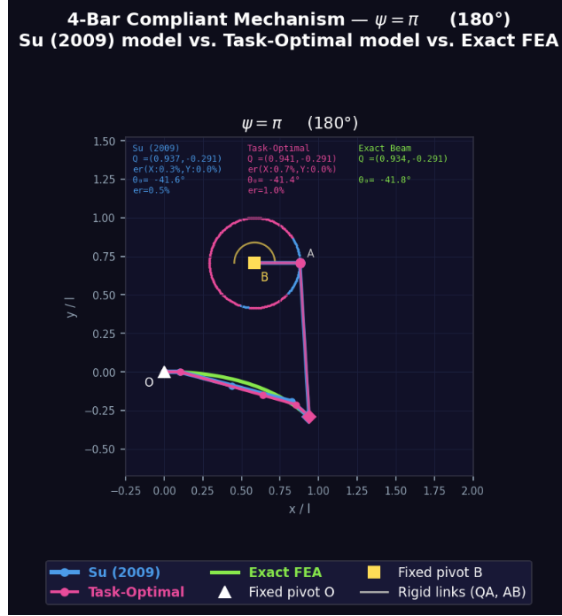


Figure 6: State of the simulation when crank angle $\psi = 180^\circ$. The locus arc is dynamically highlighted in pink where the task-optimal model outperforms the global model, and blue where the global model is superior.

6 Interactive Application

To visually communicate the discrepancy between the analytical approximations and the exact continuous solution, a custom interactive Python GUI was developed. The application simultaneously animates the real-time deformation of the exact BVP continuum beam alongside the rigid link chains of both the Su (2009) global PRB model and the newly proposed task-optimal PRB model.

The application calculates and continuously displays the coordinates, tip angles, and the dynamic percent error of the PRB models against the ground truth. To immediately convey model efficacy, the locus path of the crank is dynamically painted during the animation: the path is highlighted in pink when the task-optimal model traces closer to the exact solution, and blue when the Su model is closer.

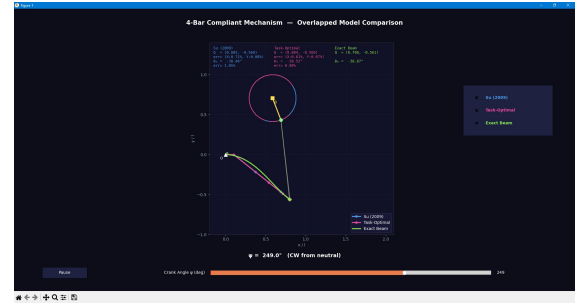


Figure 7: GUI of the custom interactive application used to visualize the dynamic tracking errors of both PRB models against the exact continuous beam.

7 Conclusions

A 3R PRB model is highly desirable for compliant mechanism design because it replaces complex continuum calculus with straightforward rigid-body kinematics. Su (2009) proposed an elegant, globally optimized formulation that performs remarkably well under general loading conditions.

However, this paper demonstrates that for mechanisms subjected to highly coupled, non-linear geometric constraints (such as a 4-bar linkage), a task-specific optimization strategy yields vastly superior results. By exclusively training the evolutionary algorithm on the exact kinematic trajectory of the target mechanism, the PRB model achieved a $4,400\times$ reduction in tracking error. While this task-specific model sacrifices accuracy under arbitrary baseline loads, its near-perfect fidelity within its operational domain highlights the immense value of application-specific optimization in advanced compliant mechanism design.

References

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